Minimum Passes Of Matrix

AlgoExpert · Medium · Graphs

The problem

- Each pass, every negative adjacent to a positive flips sign.
- Return the number of passes to clear all negatives, or -1 if impossible.
- Multi-source BFS by levels.

Intuition

- Multi-source BFS: each pass, positives flip their negative neighbours.

Brute force vs optimal

- Simulate passes by rescanning the whole matrix → $O((w \cdot h)^2)$
- Multi-source BFS: one level per pass → $O(w \cdot h)$
- Zeros block propagation

Approach

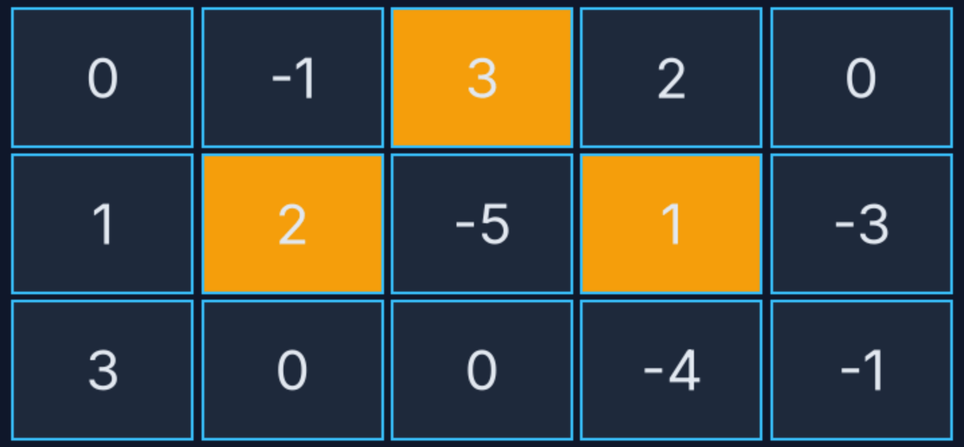
- 1. Queue all positives
- 2. Process one level per pass
- 3. Flip negative neighbours and enqueue them
- 4. If negatives remain, return minus one

Worked example

0	-1	-3	2	0
1	-2	-5	-1	-3
3	0	0	-4	-1

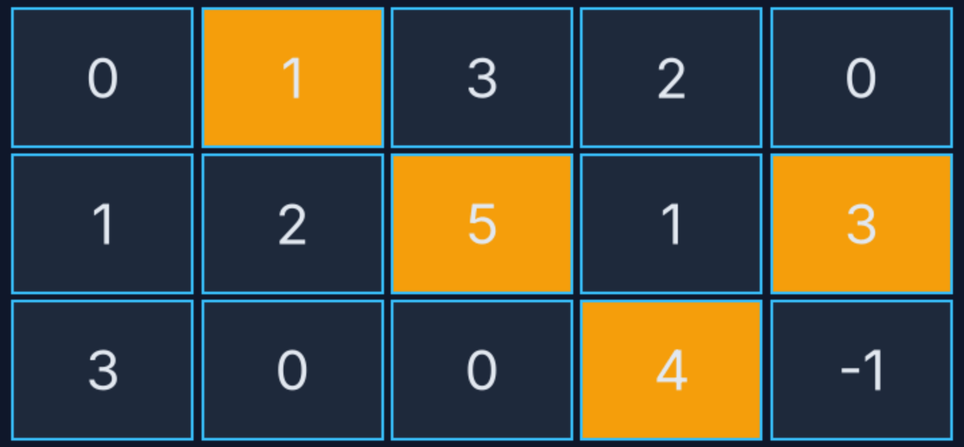
Start: all positives in the queue

Worked example



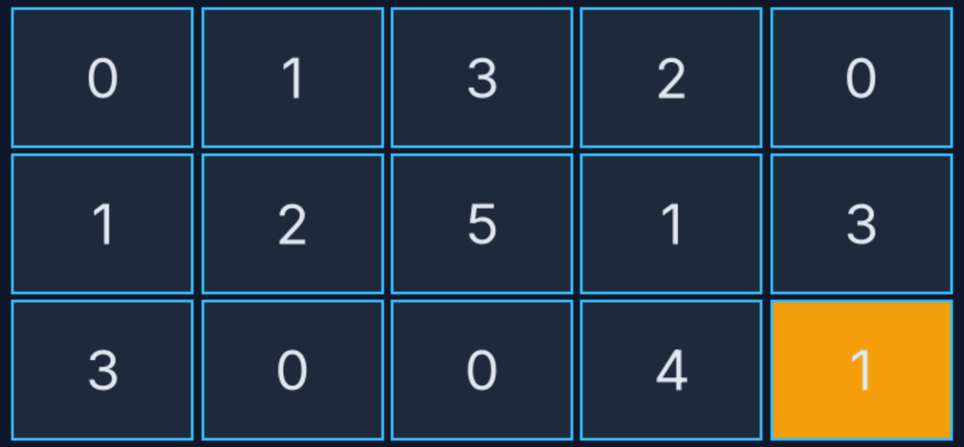
Pass 1: 3 cell(s) flipped

Worked example



Pass 2: 4 cell(s) flipped

Worked example



Pass 3: 1 cell(s) flipped

Worked example

0	1	3	2	0
1	2	5	1	3
3	0	0	4	1

No negatives left → 3 passes

Java solution

```java
public class MinimumPassesOfMatrix {
    public static int minimumPassesOfMatrix(int[][] m) {
        ArrayDeque<int[]> q = new ArrayDeque<>();
        for (int r = 0; r < m.length; r++) for (int c = 0; c < m[0].length; c++) if (m[r][c] > 0) q.add(new int[]{r, c});
        int passes = 0;
        int[][] dirs = {{1, 0}, {-1, 0}, {0, 1}, {0, -1}};
        while (!q.isEmpty()) {
            int size = q.size();
            boolean flipped = false;
            for (int i = 0; i < size; i++) {
                int[] p = q.poll();
                for (int[] d : dirs) {
                    int r = p[0] + d[0], c = p[1] + d[1];
                    if (r >= 0 && c >= 0 && r < m.length && c < m[0].length && m[r][c] < 0) {
                        m[r][c] = -m[r][c]; q.add(new int[]{r, c}); flipped = true;
                    }
                }
            }
            if (flipped) passes++;
        }
        for (int[] row : m) for (int v : row) if (v < 0) return -1;
        return passes;
    }
```

Complexity

- Time: O(w*h)
- Space: O(w*h)

Edge cases & pitfalls

- Negatives unreachable (walled by zeros) → −1
- No negatives at all → 0
- Only count passes that flip something

Interview tips & follow-ups

- Same pattern: Rotting Oranges
- Explain why BFS levels = passes

Recap

- Multi-source BFS: each pass, positives flip their negative neighbours.
- Complexity: O(w*h)